

Basic Stochastic Processes

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CHAPTER 1

Martingale Inequalities and Convergence

1. Doob's Martingale Inequalities

Proposition 1.1 (Doob's maximal inequality). *Suppose that $\xi_n, n \in \mathbb{N}$, is a non-negative submartingale (with respect to a filtration $\mathcal{F}_n$). Then for any $\lambda > 0$*

$$\lambda P \left(\max_{k \leq n} \xi_k \geq \lambda \right) \leq E \left(\xi_n 1_{\{\max_{k \leq n} \xi_k \geq \lambda\}} \right),$$

where 1_A is the characteristic function of a set A .

PROOF. We put $\xi_n^* = \max_{k \leq n} \xi_k$ for brevity. For $\lambda > 0$ let us define

$$\tau = \min \{k \leq n : \xi_k \geq \lambda\},$$

if there is a $k \leq n$ such that $\xi_k \geq \lambda$, and $\tau = n$ otherwise. Then τ is a stopping time such that $\tau \leq n$ a.s. Since ξ_n is a submartingale,

$$E(\xi_n) \geq E(\xi_\tau).$$

But

$$E(\xi_\tau) = E(\xi_\tau 1_{\{\xi_n^* \geq \lambda\}}) + E(\xi_\tau 1_{\{\xi_n^* < \lambda\}}).$$

Observe that if $\xi_n^* \geq \lambda$, then $\xi_\tau \geq \lambda$. Moreover, if $\xi_n^* < \lambda$, then $\tau = n$, and so $\xi_\tau = \xi_n$. Therefore

$$E(\xi_n) \geq E(\xi_\tau) \geq \lambda P(\xi_n^* \geq \lambda) + E(\xi_n 1_{\{\xi_n^* < \lambda\}}),$$

It follows that

$$\lambda P(\xi_n^* \geq \lambda) \leq E(\xi_n) - E(\xi_n 1_{\{\xi_n^* < \lambda\}}) = E(\xi_n 1_{\{\xi_n^* \geq \lambda\}}),$$

completing the proof. □

Theorem 1.2 (Doob's maximal L^2 inequality). *If $\xi_n, n \in \mathbb{N}$, is a non-negative square integrable submartingale (with respect to a filtration $\mathcal{F}_n$), then*

$$(4.2) \quad E \left| \max_{k \leq n} \xi_k \right|^2 \leq 4E|\xi_n|^2.$$

PROOF. Put $\xi_n^* = \max_{k \leq n} \xi_k$. By Exercise 1.9, Proposition ??, the Fubini theorem and finally the Cauchy-Schwarz inequality

$$\begin{aligned} E|\xi_n^*|^2 &= 2 \int_0^\infty t P(\xi_n^* > t) dt \leq 2 \int_0^\infty E(\xi_n 1_{\{\xi_n^* \geq t\}}) dt \\ &= 2 \int_0^\infty \left(\int_{\{\xi_n^* \geq t\}} \xi_n dP \right) dt = 2 \int_\Omega \xi_n \left(\int_0^{\xi_n^*} dt \right) dP \\ &= 2 \int_\Omega \xi_n \xi_n^* dP = 2E(\xi_n \xi_n^*) \leq 2(E|\xi_n|^2)^{1/2} (E|\xi_n^*|^2)^{1/2}. \end{aligned}$$

Dividing by $(E|\xi_n|^2)^{1/2}$, we get (4.2). □

The proof of Doob's Convergence Theorem in the next section hinges on an inequality involving the number of upcrossings.

DEFINITION 1.3. Given an adapted sequence of random variables $\xi_1, \xi_2, \dots$ and two real numbers $a < b$, we define a gambling strategy $\alpha_1, \alpha_2, \dots$ by putting

$$\alpha_1 = 0$$

and for $n = 1, 2, \dots$

$$\alpha_{n+1} = \begin{cases} 1 & \text{if } \alpha_n = 0 \text{ and } \xi_n < a, \\ 1 & \text{if } \alpha_n = 1 \text{ and } \xi_n \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

It will be called the **upcrossings strategy**. Each $k = 1, 2, \dots$ such that $\alpha_k = 1$ and $\alpha_{k+1} = 0$ will be called an upcrossing of the interval $[a, b]$. The upcrossings form a (finite or infinite) increasing sequence

$$u_1 < u_2 < \dots$$

The number of upcrossings made up to time n , that is, the largest k such that $u_k \leq n$ will be denoted by $U_n[a, b]$ (we put $U_n[a, b] = 0$ if no such k exists).

The meaning of the above definition is this. Initially, we refrain from playing the game and wait until ξ_n becomes less than a . As soon as this happens, we start playing unit stakes at each round of the game and continue until ξ_n becomes greater than b . At this stage we refrain from playing again, wait until ξ_n becomes less than a , and so on. The strategy α_n is defined in such a way that $\alpha_n = 0$ whenever we refrain from playing the n -th game, and $\alpha_n = 1$ otherwise. During each run of consecutive games with $\alpha_n = 1$ the process ξ_n crosses the interval $[a, b]$, starting below a and finishing above b . This is what is meant by an upcrossing. Observe that each upcrossing will increase our total winnings by at least $b - a$. For convenience, we identify each upcrossing with its last step k , such that $\alpha_k = 1$ and $\alpha_{k+1} = 0$. A typical sample path of the upcrossings strategy is shown in Figure ??.

FIGURE 1. Typical paths of ξ_n , α_n and ζ_n ; upcrossings are indicated by bold lines

EXERCISE 1.1. Verify that the upcrossings strategy α_n is indeed a gambling strategy.

Hint: You want to prove that α_n is $\mathcal{F}_{n-1}$ -measurable for each n . Since the upcrossings strategy is defined by induction, a proof by induction on n may be your best bet.

Lemma 1.4 (Upcrossings Inequality). If $\xi_1, \xi_2, \dots$ is a supermartingale and $a < b$, then

$$(b - a)E(U_n[a, b]) \leq E((\xi_n - a)^-).$$

By x^- we denote the negative part of a real number x , i.e. $x^- = \max\{0, -x\}$.

PROOF. Let

$$\zeta_n = \alpha_1(\xi_1 - \xi_0) + \dots + \alpha_n(\xi_n - \xi_{n-1})$$

be the total winnings at step $n = 1, 2, \dots$ if the upcrossings strategy is followed, see Figure ??. It will be convenient to put $\zeta_0 = 0$. By Proposition 3.1 (one cannot beat the system using a gambling strategy) ζ_n is a supermartingale.

Let us fix an n and put $k = U_n[a, b]$, so that

$$0 < u_1 < u_2 < \dots < u_k \leq n.$$

Clearly, each upcrossing increases the total winnings by $b - a$,

$$\zeta_{u_i} - \zeta_{u_{i-1}} \geq b - a$$

for $i = 1, \dots, k$. (We put $u_0 = 0$ for simplicity.) Moreover,

$$\zeta_n - \zeta_{u_k} \geq -(\xi_n - a)^-.$$

It follows that

$$\zeta_n \geq (b - a)U_n[a, b] - (\xi_n - a)^-.$$

Taking the expectation on both sides, we get

$$E(\zeta_n) \geq (b - a)E(U_n[a, b]) - E((\xi_n - a)^-).$$

But ζ_n is a supermartingale, so

$$0 = E(\zeta_1) \geq E(\zeta_n),$$

which proves the Upcrossings Inequality. $\square$

2. Doob's Martingale Convergence Theorem

Theorem 1.5 (Doob's Martingale Convergence Theorem). *Suppose that $\xi_1, \xi_2, \dots$ is a supermartingale (with respect to a filtration $\mathcal{F}_1, \mathcal{F}_2, \dots$) such that*

$$\sup_n E(|\xi_n|) < \infty.$$

Then there is an integrable random variable ξ such that

$$\lim_{n \rightarrow \infty} \xi_n = \xi \quad \text{a.s.}$$

REMARK 1.1. *In particular, the theorem is valid for martingales because every martingale is a supermartingale. It is also valid for submartingales, since ξ_n is a submartingale if and only if $-\xi_n$ is a supermartingale.*

REMARK 1.2. *Observe that even though all the ξ_n as well as the limit ξ are integrable random variables, it is claimed only that ξ_n trends to ξ a.s. Note that no convergence in L^1 is asserted.*

PROOF (Proof of Doob's Martingale Convergence Theorem). *By the Upcrossings Inequality (Lemma ??)*

$$E(U_n[a, b]) \leq \frac{E((\xi_n - a)^-)}{b - a} \leq \frac{M + |a|}{b - a} < \infty,$$

where

$$M = \sup_n E(|\xi_n|) < \infty.$$

Since $U_n[a, b]$ is a non-decreasing sequence, it follows that

$$E\left(\lim_{n \rightarrow \infty} U_n[a, b]\right) = \lim_{n \rightarrow \infty} E(U_n[a, b]) \leq \frac{M + |a|}{b - a} < \infty.$$

This implies that

$$P\left\{\lim_{n \rightarrow \infty} U_n[a, b] < \infty\right\} = 1$$

for any $a < b$. Since the set of all pairs of rational numbers $a < b$ is countable, the event

$$(4.3) \quad A = \bigcap_{a < b \text{ rational}} \left\{\lim_{n \rightarrow \infty} U_n[a, b] < \infty\right\}$$

has probability 1. (The intersection of countably many events has probability 1 if each of these events has probability 1.)

We claim that the sequence ξ_n converges a.s. to a limit ξ . Consider the set

$$B = \left\{ \liminf_n \xi_n < \limsup_n \xi_n \right\} \subset \Omega$$

on which the sequence ξ_n fails to converge. Then for any $\omega \in B$ there are rational numbers a, b such that

$$\liminf_n \xi_n(\omega) < a < b < \limsup_n \xi_n(\omega),$$

implying that $\lim_{n \rightarrow \infty} U_n[a, b](\omega) = \infty$. This means that B and the event A in (4.3) are disjoint, so $P(B) = 0$, since $P(A) = 1$, which proves the claim.

It remains to show that the limit ξ is an integrable random variable. By Fatou's lemma

$$\begin{aligned} E(|\xi|) &= E(\liminf_n |\xi_n|) \\ &\leq \liminf_n E(|\xi_n|) \\ &< \sup_n E(|\xi_n|) < \infty. \end{aligned}$$

This completes the proof. □

EXERCISE 1.2. Show that if ξ_n is a non-negative supermartingale, then it converges a.s. to an integrable random variable.

Hint: To apply Doob's Theorem all you need to verify is that the sequence ξ_n is bounded in L^1 , i.e. the supremum of $E(|\xi_n|)$ is less than ∞ .

3. Uniform Integrability and L^1 Convergence of Martingales

The conditions of Doob's theorem imply pointwise (a.s.) convergence of martingales. In this section we shall study convergence in L^1 . To this end we introduce a stronger condition called uniform integrability. Proposition ?? shows that it is a necessary condition for L^1 convergence. In Theorem ?? we prove that uniform integrability is in fact sufficient for a martingale to converge in L^1 . This enables us to show that each integrable martingale is of the form $E(\xi|\mathcal{F}_n)$. As an application we give a martingale proof of Kolmogorov's 0-1 law.

EXERCISE 1.3. Show that a random variable ξ is integrable if and only if for every $\varepsilon > 0$ there exists an $M > 0$ such that

$$\int_{\{|\xi| > M\}} |\xi| dP < \varepsilon.$$

Hint: Split Ω into two sets: $\{|\xi| > M\}$ and $\{|\xi| \leq M\}$. The integrals of $|\xi|$ over these sets must add up to $E(|\xi|)$. As M increases, one of the integrals increases, while the other one decreases. Investigate their limits as $M \rightarrow \infty$.

Thus, for any sequence ξ_n of integrable random variables and any $\varepsilon > 0$ there is a sequence of numbers $M_n > 0$ such that

$$\int_{\{|\xi_n| > M_n\}} |\xi_n| dP < \varepsilon.$$

If the M_n are independent of n , then we say that the sequence ξ_n is uniformly integrable.

DEFINITION 1.6. A sequence $\xi_1, \xi_2, \dots$ of random variables is called **uniformly integrable** if for every $\varepsilon > 0$ there exists an $M > 0$ such that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < \varepsilon$$

for all $n = 1, 2, \dots$

EXERCISE 1.4. Let $\Omega = [0, 1]$ with the σ -field of Borel sets and Lebesgue measure. Take

$$\xi_n = n1_{(0, \frac{1}{n})}.$$

Show that the sequence $\xi_1, \xi_2, \dots$ is not uniformly integrable.

Hint: What is the integral of ξ_n over $\{\xi_n > M\}$ if $n > M$?

Proposition 1.7. Uniform integrability is a necessary condition for a sequence $\xi_1, \xi_2, \dots$ of integrable random variables to converge in L^1 .

Lemma 1.8. If ξ is integrable, then for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon.$$

PROOF (Proof of Lemma ??). Let $\varepsilon > 0$. Since ξ is integrable, by Exercise ?? there is an $M > 0$ such that

$$\int_{\{|\xi| > M\}} |\xi| dP < \frac{\varepsilon}{2}.$$

Now

$$\begin{aligned} \int_A |\xi| dP &= \int_{A \cap \{|\xi| \leq M\}} |\xi| dP + \int_{A \cap \{|\xi| > M\}} |\xi| dP \\ &\leq \int_A M dP + \int_{\{|\xi| > M\}} |\xi| dP \\ &< MP(A) + \frac{\varepsilon}{2}. \end{aligned}$$

Let $\delta = \frac{\varepsilon}{2M}$. Then

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon,$$

as required. □

EXERCISE 1.5. Let ξ be an integrable random variable and $\mathcal{F}_1, \mathcal{F}_2, \dots$ a filtration. Show that $E(\xi | \mathcal{F}_n)$ is a uniformly integrable martingale.

Hint: Use Lemma ??.

PROOF (Proof of Proposition ??). Suppose that $\xi_n \rightarrow \xi$ in L^1 , i.e. $E|\xi_n - \xi| \rightarrow 0$. We take any $\varepsilon > 0$. There is an integer N such that

$$n \geq N \implies E|\xi_n - \xi| < \frac{\varepsilon}{2}.$$

By Lemma ?? there is a $\delta > 0$ such that

$$P(A) < \delta \implies \int_A |\xi| dP < \frac{\varepsilon}{2}.$$

Taking a smaller $\delta > 0$ if necessary, we also have

$$P(A) < \delta \implies \int_A |\xi_n| dP < \varepsilon \quad \text{for } n = 1, \dots, N.$$

We claim that there is an $M > 0$ such that

$$P\{|\xi_n| > M\} < \delta$$

for all n . Indeed, since

$$E(|\xi_n|) \geq \int_{\{|\xi_n| > M\}} |\xi_n| dP \geq MP\{|\xi_n| > M\},$$

it suffices to take

$$M = \frac{1}{\delta} \sup_n E(|\xi_n|).$$

(Because the sequence ξ_n converges in L^1 , it is bounded in L^1 , so the supremum is $< \infty$.)

Now, since $P\{|\xi_n| > M\} < \delta$,

$$\begin{aligned} \int_{\{|\xi_n| > M\}} |\xi_n| dP &\leq \int_{\{|\xi_n| > M\}} |\xi| dP + \int_{\{|\xi_n| > M\}} |\xi_n - \xi| dP \\ &\leq \int_{\{|\xi_n| > M\}} |\xi| dP + E(|\xi_n - \xi|) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

for any $n > N$ and

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < \varepsilon$$

for any $n = 1, \dots, N$, completing the proof.

EXERCISE 1.6. Show that a uniformly integrable sequence of random variables is bounded in L^1 , i.e.

$$\sup_n E(|\xi_n|) < \infty.$$

Hint: Write $E(|\xi_n|)$ as the sum of the integrals of $|\xi_n|$ over $\{|\xi_n| > M\}$ and $\{|\xi_n| \leq M\}$.

Exercise ?? implies that each uniformly integrable martingale satisfies the conditions of Doob's theorem. Therefore it converges a.s. to an integrable random variable. We shall show that in fact it converges in L^1 .

Theorem 1.9. Every uniformly integrable supermartingale (submartingale) ξ_n converges in L^1 .

PROOF. By Exercise ?? the sequence ξ_n is bounded in L^1 , so it satisfies the conditions of Theorem ?? (Doob's Martingale Convergence Theorem). Therefore, there is an integrable random variable ξ such that $\xi_n \rightarrow \xi$ a.s. We can assume without loss of generality that $\xi = 0$ (since $\xi_n - \xi$ can be taken in place of ξ_n). That is to say,

$$P\left\{\lim_n \xi_n = 0\right\} = 1.$$

It follows that $\xi_n \rightarrow 0$ in probability, i.e. for any $\varepsilon > 0$

$$P\{|\xi_n| > \varepsilon\} \rightarrow 0$$

as $n \rightarrow \infty$. This is because by Fatou's lemma

$$\begin{aligned} \limsup_n P\{|\xi_n| > \varepsilon\} &\leq P(\limsup_n \{|\xi_n| > \varepsilon\}) \\ &\leq P(\Omega \setminus \{\lim_n \xi_n = 0\}) \\ &= 0. \end{aligned}$$

Let $\varepsilon > 0$. By uniform integrability there is an $M > 0$ such that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP \leq \frac{\varepsilon}{3}$$

for all n . Since $\xi_n \rightarrow 0$ in probability, there is an integer N such that if $n > N$, then

$$P\left\{|\xi_n| > \frac{\varepsilon}{3}\right\} < \frac{\varepsilon}{3M}.$$

We can assume without loss of generality that $M > \frac{\varepsilon}{3}$. Then

$$\begin{aligned} E(|\xi_n|) &= \int_{\{|\xi_n| > M\}} |\xi_n| dP + \int_{\{M \geq |\xi_n| > \frac{\varepsilon}{3}\}} |\xi_n| dP \\ &\quad + \int_{\{\frac{\varepsilon}{3} \geq |\xi_n|\}} |\xi_n| dP \\ &\leq \frac{\varepsilon}{3} + MP\left\{|\xi_n| > \frac{\varepsilon}{3}\right\} + \frac{\varepsilon}{3}P\left\{\frac{\varepsilon}{3} \geq |\xi_n|\right\} \\ &< \varepsilon \end{aligned}$$

for all $n > N$. This proves that $E(|\xi_n|) \rightarrow 0$, that is, $\xi_n \rightarrow 0$ in L^1 . $\square$

Theorem 1.10. Let ξ_n be a uniformly integrable martingale. Then

$$\xi_n = E(\xi | \mathcal{F}_n),$$

where $\xi = \lim_n \xi_n$ is the limit of ξ_n in L^1 and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ is the filtration generated by ξ_n .

PROOF. For any $m > n$

$$E(\xi_m | \mathcal{F}_n) = \xi_n,$$

i.e. for any $A \in \mathcal{F}_n$

$$\int_A \xi_m dP = \int_A \xi_n dP.$$

Let n be an arbitrary integer and let $A \in \mathcal{F}_n$. For any $m > n$

$$\begin{aligned} \left| \int_A (\xi_n - \xi) dP \right| &= \left| \int_A (\xi_m - \xi) dP \right| \\ &\leq \int_A |\xi_m - \xi| dP \\ &\leq E(|\xi_m - \xi|) \rightarrow 0 \end{aligned}$$

as $m \rightarrow \infty$. It follows that

$$\int_A \xi_n dP = \int_A \xi dP$$

for any $A \in \mathcal{F}_n$, so $\xi_n = E(\xi | \mathcal{F}_n)$. $\square$

EXERCISE 1.7. Show that if ξ_n is a martingale and $\xi_n \rightarrow a$ in L^1 for some $a \in \mathbb{R}$, then $\xi_n = a$ a.s. for each n .

Hint: Apply Theorem ??.

Theorem 1.11 (Kolmogorov's 0–1 Law). Let $\eta_1, \eta_2, \dots$ be a sequence of independent random variables. We define the tail σ -field

$$\mathcal{T} = \mathcal{T}_1 \cap \mathcal{T}_2 \cap \dots,$$

where $\mathcal{T}_n = \sigma(\eta_n, \eta_{n+1}, \dots)$. Then

$$P(A) = 0 \text{ or } 1$$

for any $A \in \mathcal{T}$.

PROOF. Take any $A \in \mathcal{T}$ and define

$$\xi_n = E(1_A | \mathcal{F}_n),$$

where $\mathcal{F}_n = \sigma(\eta_1, \dots, \eta_n)$. By Exercise ?? ξ_n is a uniformly integrable martingale, so $\xi_n \rightarrow \xi$ in L^1 . By Theorem ??

$$E(\xi | \mathcal{F}_n) = E(1_A | \mathcal{F}_n)$$

for all n . Both $\xi = \lim_n \xi_n$ and 1_A are measurable with respect to the σ -field

$$\mathcal{F}_\infty = \sigma(\eta_1, \eta_2, \dots).$$

The family $\mathcal{G}$ consisting of all sets $B \in \mathcal{F}$ such that $\int_B \xi dP = \int_B 1_A dP$ is a σ -field containing $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \dots$. As a result, $\mathcal{G}$ contains the σ -field $\mathcal{F}_\infty$ generated by the family $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \dots$. By Lemma 2.1 it follows that $\xi = 1_A$ a.s.

Since η_n is a sequence of independent random variables, the σ -fields $\mathcal{F}_n$ and $\mathcal{T}_{n+1}$ are independent. Because $\mathcal{T} \subset \mathcal{T}_{n+1}$, the σ -fields $\mathcal{F}_n$ and $\mathcal{T}$ are independent. Being $\mathcal{T}$ -measurable, 1_A is therefore independent of $\mathcal{F}_n$ for any n . This means that

$$\xi_n = E(1_A | \mathcal{F}_n) = E(1_A) = P(A) \quad \text{a.s.}$$

Therefore the limit $\lim_{n \rightarrow \infty} \xi_n = \xi$ is also constant and equal to $P(A)$ a.s. This means that $P(A) = 1_A$ a.s., so $P(A) = 0$ or 1 . $\square$

EXERCISE 1.8. Show that if $A_n \in \sigma(\xi_n)$ for each n , then the events

$$\limsup_n A_n = \bigcap_{j \geq 1} \bigcup_{i \geq j} A_i$$

and

$$\liminf_n A_n = \bigcup_{j \geq 1} \bigcap_{i \geq j} A_i$$

belong to the tail σ -field $\mathcal{T}$.

Hint: You need to write $\limsup_n A_n$ and $\liminf_n A_n$ in terms of $A_k, A_{k+1}, \dots$ for any k , that is, to show that $\limsup_n A_n$ and $\liminf_n A_n$ will not be affected if any finite number of sets are removed from the sequence $A_1, A_2, \dots$

EXERCISE 1.9. Use Kolmogorov's 0-1 law to show that in a sequence of coin tosses there are a.s. infinitely many heads.

Hint: Show that the event

$$\{\eta_1, \eta_2, \dots \text{ contains infinitely many heads} \}$$

belongs to the σ -field $\mathcal{T}$. Can the probability of this event be 0? The probability of the event

$$\{\eta_1, \eta_2, \dots \text{ contains infinitely many tails} \}$$

should be the same. Can both probabilities be equal to 0? Can you simultaneously have finitely many heads and finitely many tails in the sequence $\eta_1, \eta_2, \dots$?

4. Solutions

SOLUTION 1.1. Because $\alpha_1 = 0$ is constant, it is $\mathcal{F}_0 = \{\emptyset, \Omega\}$ -measurable. Suppose that α_n is $\mathcal{F}_{n-1}$ -measurable for some $n = 1, 2, \dots$. Then

$$\{\alpha_n = 0, \xi_n < a\} \cup \{\alpha_n = 1, \xi_n \leq b\} \in \mathcal{F}_n$$

because $\mathcal{F}_{n-1} \subset \mathcal{F}_n$ and ξ_n is $\mathcal{F}_n$ -measurable. This means that

$$\alpha_{n+1} = 1_{\{\alpha_n=0, \xi_n < a\} \cup \{\alpha_n=1, \xi_n \leq b\}}$$

is $\mathcal{F}_n$ -measurable. By induction it follows that α_n is $\mathcal{F}_{n-1}$ -measurable for each $n = 1, 2, \dots$, so $\alpha_1, \alpha_2, \dots$ is a gambling strategy.

SOLUTION 1.2. For a non-negative supermartingale

$$\sup_n E(|\xi_n|) = \sup_n E(\xi_n) \leq E(\xi_1) = E(|\xi_1|) < \infty,$$

since

$$E(\xi_n) \leq E(\xi_1)$$

for each $n = 1, 2, \dots$. Thus Doob's Martingale Convergence Theorem implies that ξ_n converges a.s. to an integrable limit.

SOLUTION 1.3. Necessity. Suppose that ξ is integrable. It follows that

$$P\{|\xi| < \infty\} = 1.$$

The sequence of random variables $|\xi|1_{\{|\xi|>M\}}$ indexed by $M = 1, 2, \dots$ is monotone and

$$|\xi|1_{\{|\xi|>M\}} \searrow 0 \quad \text{as } M \rightarrow \infty$$

on the set $\{|\xi| < \infty\}$, i.e. a.s. By the monotone convergence theorem for integrals

$$\int_{\{|\xi|>M\}} |\xi| dP \searrow 0 \quad \text{as } M \rightarrow \infty.$$

It follows that for every $\varepsilon > 0$ there exists an $M > 0$ such that

$$\int_{\{|\xi|>M\}} |\xi| dP < \varepsilon.$$

Sufficiency. Take $\varepsilon = 1$. There exists an $M > 0$ such that

$$\int_{\{|\xi|>M\}} |\xi| dP < 1.$$

Then

$$\begin{aligned} E(|\xi|) &= \int_{\Omega} |\xi| dP \\ &= \int_{\{|\xi|>M\}} |\xi| dP + \int_{\{|\xi|\leq M\}} |\xi| dP \\ &< 1 + MP\{|\xi| \leq M\} \\ &\leq 1 + M < \infty. \end{aligned}$$

SOLUTION 1.4. For any $M > 0$ and any $n > M$ we have

$$(0, \frac{1}{n}) = \{\xi_n > M\},$$

so

$$\int_{\{\xi_n > M\}} \xi_n dP = \int_{(0, \frac{1}{n})} ndP = 1.$$

This means that there is no $M > 0$ such that for all n

$$\int_{\{\xi_n > M\}} \xi_n dP < \frac{1}{2}.$$

The sequence ξ_n is not uniformly integrable.

SOLUTION 1.5. In Example 3.4 it was verified that $\xi_n = E(\xi|\mathcal{F}_n)$ is a martingale. Let $\varepsilon > 0$. By Lemma ?? there is a $\delta > 0$ such that

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon.$$

By Jensen's inequality $|\xi_n| \leq E(|\xi||\mathcal{F}_n)$ a.s., so

$$E(|\xi|) \geq E(|\xi_n|) \geq \int_{\{|\xi_n| \geq M\}} |\xi_n| dP \geq MP\{|\xi_n| > M\}.$$

If we take $M > E(|\xi|)/\delta$, then

$$P\{|\xi_n| > M\} < \delta.$$

Since $\{|\xi_n| > M\} \in \mathcal{F}_n$, it follows that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP \leq \int_{\{|\xi_n| > M\}} E(|\xi| | \mathcal{F}_n) dP = \int_{\{|\xi_n| > M\}} |\xi| dP < \varepsilon,$$

proving that $\xi_n = E(\xi | \mathcal{F}_n)$ is a uniformly integrable sequence.

SOLUTION 1.6. Because ξ_n is a uniformly integrable sequence, there is an $M > 0$ such that for all n

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < 1.$$

It follows that

$$\begin{aligned} E(|\xi_n|) &= \int_{\{|\xi_n| > M\}} |\xi_n| dP + \int_{\{|\xi_n| \leq M\}} |\xi_n| dP \\ &< 1 + MP\{|\xi_n| \leq M\} \\ &< 1 + M < \infty \end{aligned}$$

for all n , proving that ξ_n is a bounded sequence in L^1 .

SOLUTION 1.7. By Theorem ??, $\xi_n = E(a | \mathcal{F}_n)$ a.s. But $E(a | \mathcal{F}_n) = a$ a.s., which proves that $\xi_n = a$ a.s.

SOLUTION 1.8. Observe that

$$\limsup_n A_n = \bigcap_{j \geq k} \bigcup_{i \geq j} A_i$$

for any k . Since

$$A_i \in \sigma(\xi_i) \subset \mathcal{T}_k$$

for every $i \geq k$, it follows that

$$\limsup_n A_n = \bigcap_{j \geq k} \bigcup_{i \geq j} A_i \in \mathcal{T}_k$$

for every k . Therefore

$$\sup_n A_n \in \mathcal{T}.$$

The argument for $\liminf_n A_n$ is similar.

SOLUTION 1.9. Let $\eta_1, \eta_2, \dots$ be a sequence of coin tosses, i.e. independent random variables with values $+1, -1$ (heads or tails) taken with probability $\frac{1}{2}$ each. Consider the following event:

$$A = \{\eta_1, \eta_2, \dots \text{ contains infinitely many heads} \}.$$

This event belongs to the tail σ -field $\mathcal{T}$ because

$$A = \limsup_n A_n,$$

where

$$A_n = \{\eta_n = \text{heads}\} \in \sigma(\eta_n)$$

(see Exercise ??). Thus, by Kolmogorov's 0-1 law $P(A) = 0$ or 1 . However, it cannot be 0 because the event

$$B = \{\eta_1, \eta_2, \dots \text{ contains infinitely many tails} \}$$

has the same probability by symmetry and $\Omega = A \cup B$ (there must be infinitely many heads or infinitely many tails).

This chapter is concerned with an interesting class of sequences of random variables taking values in a finite or countable set, called the state space, and satisfying the so-called Markov property. One of the simplest examples is provided by a symmetric random walk ξ_n with values in the set of integers $\mathbb{Z}$. If ξ_n is equal to some $i \in \mathbb{Z}$ at time n , then in the next time instance $n+1$ it will jump either to $i+1$, with probability $\frac{1}{2}$, or to $i-1$, also with probability $\frac{1}{2}$. What makes this model interesting is that the value of ξ_{n+1} at time $n+1$ depends on the past only through the value at time n . This is the Markov property characterizing Markov chains. There are numerous examples of Markov chains, with a multitude of applications.

From the mathematical point of view, Markov chains are both simple and difficult. Their definition and basic properties do not involve any complicated notions or sophisticated mathematics. Yet, any deeper understanding of Markov chains requires quite advanced tools. For example, this is so for problems related to the long-time behaviour of Markov processes. In this chapter we shall try to maintain a balance between the accessibility of exposition and the depth of mathematical results. Various concepts will be introduced. In particular, we shall discuss the classification of states and its relevance to the asymptotic behaviour of transition probabilities. This will turn out to be closely linked to ergodicity and the existence and uniqueness of invariant measures.

CHAPTER 2

Markov Chains

1. First Examples and Definitions

1.1. Example 5.1. In some homes the use of the telephone can become quite a sensitive issue. Suppose that if the phone is free during some period of time, say the n th minute, then with probability p , where $0 < p < 1$, it will be busy during the next minute. If the phone has been busy during the n th minute, it will become free during the next minute with probability q , where $0 < q < 1$. Assume that the phone is free in the 0th minute. We would like to answer the following two questions.

- (1) What is the probability x_n that the telephone will be free in the n th minute?
- (2) What is $\lim_{n \rightarrow \infty} x_n$, if it exists?

Denote by A_n the event that the phone is free during the n th minute and let $B_n = \Omega \setminus A_n$ be its complement, i.e. the event that the phone is busy during the n th minute. The conditions of the example give us

$$(5.1) \quad P(B_{n+1}|A_n) = p,$$

$$(5.2) \quad P(A_{n+1}|B_n) = q.$$

We also assume that $P(A_0) = 1$, i.e. $x_0 = 1$. Using this notation, we have $x_n = P(A_n)$. Then the total probability formula, see Exercise 1.10, together with (5.1)-(5.2) imply that

$$(5.3) \quad x_{n+1} = P(A_{n+1})$$

$$(1) \quad = P(A_{n+1}|A_n)P(A_n) + P(A_{n+1}|B_n)P(B_n)$$

$$(2) \quad = (1-p)x_n + q(1-x_n) = q + (1-p-q)x_n.$$

It's a bit tricky to find an explicit formula for x_n . To do so we suppose first that the sequence $\{x_n\}$ is convergent, i.e.

$$(5.4) \quad \lim_{n \rightarrow \infty} x_n = x.$$

The elementary properties of limits and equation (5.3), i.e. $x_{n+1} = q + (1-p-q)x_n$, yield

$$(5.5) \quad x = q + (1-p-q)x.$$

The unique solution to the last equation is

$$(5.6) \quad x = \frac{q}{q+p}.$$

In particular,

$$(5.7) \quad \frac{q}{q+p} = q + (1-p-q) \frac{q}{q+p}.$$

Subtracting (5.7) from (5.3), we infer that

$$(5.8) \quad x_{n+1} - \frac{q}{q+p} = (1-p-q) \left(x_n - \frac{q}{q+p} \right).$$

Thus, $\{x_n - \frac{q}{q+p}\}$ is a geometric sequence and therefore, for all $n \in \mathbb{N}$,

$$(3) \quad x_n - \frac{q}{q+p} = (1-p-q)^n \left(x_0 - \frac{q}{q+p} \right).$$

Hence, by taking into account the initial condition $x_0 = 1$, we have

$$(5.9) \quad x_n = \frac{q}{q+p} + \left(x_0 - \frac{q}{q+p} \right) (1-p-q)^n$$

$$(4) \quad = \frac{q}{q+p} + \frac{p}{q+p} (1-p-q)^n.$$

Let us point out that although we have used the assumption (5.4) to derive (5.8), the proof of the latter is now complete. Indeed, having proven (5.9), we can show that the assumption (5.4) is indeed satisfied. This is because the conditions $0 < p, q < 1$ imply that $|1-p-q| < 1$, and so $(1-p-q)^n \rightarrow 0$ as $n \rightarrow \infty$. Thus, (5.4) holds. This provides an answer to the second part of the example, i.e. $\lim_{n \rightarrow \infty} x_n = \frac{q}{p+q}$.

EXERCISE 2.1. *In the framework of Example ??, let y_n denote the probability that the telephone is busy in the n th minute. Supposing that $y_0 = 1$, find an explicit formula for y_n and, if it exists, $\lim_{n \rightarrow \infty} y_n$.*

REMARK 2.1. *Hint:* *This exercise can be solved directly by repeating the above argument, or indirectly by using some of the results in Example ??.*

REMARK 2.2. *The formulae (5.3) and (5.9) can be written collectively in a compact form by using vector and matrix notation. First of all, since $x_n + y_n = 1$, we get*

$$(5) \quad x_{n+1} = (1-p)x_n + qy_n,$$

$$(6) \quad y_{n+1} = px_n + (1-q)y_n.$$

Hence, the matrix version takes the form

$$(7) \quad \begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix} \begin{bmatrix} x_n \\ y_n \end{bmatrix}.$$

The situation described in Example ?? is quite typical. Often the probability of a certain event at time $n+1$ depends only on what happens at time n , but not further into the past.

Example ?? provides us with a simple case of a Markov chain. See also the following definition and exercises.

DEFINITION 2.1 (Markov chain). *Suppose that S is a finite or a countable set. Suppose also that a probability space $(\Omega, \mathcal{F}, P)$ is given. An S -valued sequence of random variables $\xi_n, n \in \mathbb{N}$, is called an S -valued Markov chain or a Markov chain on S if for all $n \in \mathbb{N}$ and all $s \in S$*

$$(5.10) \quad P(\xi_{n+1} = s \mid \xi_0, \dots, \xi_n) = P(\xi_{n+1} = s \mid \xi_n).$$

Here $P(\xi_{n+1} = s \mid \xi_n)$ is the conditional probability of the event $\{\xi_{n+1} = s\}$ with respect to random variable ξ_n , or equivalently, with respect to the σ -field $\sigma(\xi_n)$ generated by ξ_n . Similarly, $P(\xi_{n+1} = s \mid \xi_0, \dots, \xi_n)$ is the conditional probability of $\{\xi_{n+1} = s\}$ with respect to the σ -field $\sigma(\xi_0, \dots, \xi_n)$ generated by the random variables $\xi_0, \dots, \xi_n$.

Property (5.10) will usually be referred to as the Markov property of the Markov chain $\xi_n, n \in \mathbb{N}$. The set S is called the state space and the elements of S are called states.

Proposition 2.2. *The model in Example ?? and Exercise ?? is a Markov chain.*

PROOF. *Let $S = \{0, 1\}$, where 0 and 1 represent the states of the phone being free or busy. First we need to construct an appropriate probability space. Let Ω be the set of all sequences $\omega_0, \omega_1, \dots$ with values in S . Let μ_0 be any probability measure on S . For example, $\mu = \delta_0$ corresponds to the case when the phone is free at time 0. We shall define P by induction. For any S -valued sequence $s_0, s_1, \dots$ we put*

$$(5.11) \quad P(\{\omega \in \Omega : \omega_0 = s_0\}) = \mu_0(\{s_0\})$$

and

$$(5.12) \quad P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n+1\})$$

$$(8) \quad = p(s_{n+1} \mid s_n) P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n\}),$$

where $p(s \mid r)$ are the entries of the 2×2 matrix

$$(9) \quad \begin{bmatrix} p(0 \mid 0) & p(0 \mid 1) \\ p(1 \mid 0) & p(1 \mid 1) \end{bmatrix} = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}.$$

It seems reasonable to expect P to be a probability measure (with respect to the trivial σ -field of all subsets of Ω). Take this for granted, check only that $P(\Omega) = 1$.

How would you define the process $\xi_n, n \in \mathbb{N}$? We shall do it in the standard way, i.e.

$$(5.13) \quad \xi_n(\omega) = \omega_n, \omega \in \Omega.$$

First we shall show that the transition probabilities of ξ_n are what they should be, i.e.

$$(5.14) \quad P(\xi_{n+1} = 1 \mid \xi_n = 0) = p,$$

$$(5.15) \quad P(\xi_{n+1} = 0 \mid \xi_n = 1) = q.$$

The definition of conditional probability yields

$$(10) \quad P(\xi_{n+1} = 1 \mid \xi_n = 0) = \frac{P(\xi_{n+1} = 1, \xi_n = 0)}{P(\xi_n = 0)}.$$

Next, the definition of P gives

$$(11) \quad \begin{aligned} P(\xi_{n+1} = 1, \xi_n = 0) &= P(\{\omega \in \Omega : \omega_n = 0, \omega_{n+1} = 1\}) \\ &= \sum_{s_0, \dots, s_{n-1} \in S} P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n-1, \omega_n = 0, \omega_{n+1} = 1\}) \\ &= \sum_{s_0, \dots, s_{n-1} \in S} pP(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n-1, \omega_n = 0\}) \\ &= pP(\xi_n = 0), \end{aligned}$$

by (5.12). We have proven (5.14). Moreover, (5.15) follows by the same argument. A similar line of reasoning shows that ξ_n is indeed a Markov chain. For this we need to verify that for any $n \in \mathbb{N}$ and any $s_0, s_1, \dots, s_{n+1} \in S$

$$(12) \quad P(\xi_{n+1} = s_{n+1} \mid \xi_0 = s_0, \dots, \xi_n = s_n) = P(\xi_{n+1} = s_{n+1} \mid \xi_n = s_n).$$

We have

$$(13) \quad \begin{aligned} &P(\xi_0 = s_0, \dots, \xi_n = s_n, \xi_{n+1} = s_{n+1}) \\ &= P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n+1\}) \\ &= p(s_{n+1} \mid s_n)P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n\}) \\ &= p(s_{n+1} \mid s_n)P(\xi_0 = s_0, \dots, \xi_n = s_n) \end{aligned}$$

which, in view of the definition of conditional probability, gives

$$(14) \quad P(\xi_{n+1} = s_{n+1} \mid \xi_0 = s_0, \dots, \xi_n = s_n) = p(s_{n+1} \mid s_n).$$

On the other hand, by an easy generalization of (5.14) and (5.15)

$$(15) \quad P(\xi_{n+1} = s_{n+1} \mid \xi_n = s_n) = p(s_{n+1} \mid s_n),$$

which proves (5.10). □

In Example ?? the transition probabilities from state i to state j do not depend on the time n . This is an important class of Markov chains.

DEFINITION 2.3 (Time-homogeneous Markov chain). *An S -valued Markov chain $\xi_n, n \in \mathbb{N}$, is called time-homogeneous or homogeneous if for all $n \in \mathbb{N}$ and all $i, j \in S$*

$$(5.16) \quad P(\xi_{n+1} = j \mid \xi_n = i) = P(\xi_1 = j \mid \xi_0 = i).$$

The number $P(\xi_1 = j \mid \xi_0 = i)$ is denoted by $p(j \mid i)$ and called the transition probability from state i to state j . The matrix $P = [p(j \mid i)]_{j,i \in S}$ is called the transition matrix of the chain ξ_n .

EXERCISE 2.2. In the discussion so far we have seen an example of a transition matrix, $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$. Obviously the sum of the entries in each column of P is equal to 1. Prove that this is true in general.

REMARK 2.3. **Hint:** Remember that $P(\Omega \mid A) = 1$ for any event A .

DEFINITION 2.4 (Stochastic matrix). $A = [a_{ji}]_{i,j \in S}$ is called a stochastic matrix if

- (1) $a_{ji} \geq 0$, for all $i, j \in S$;
- (2) the sum of the entries in each column is 1, i.e. $\sum_{j \in S} a_{ji} = 1$ for any $i \in S$.

A is called a double stochastic matrix if both A and its transpose A^t are stochastic matrices.

Proposition 2.5. Show that a stochastic matrix is doubly stochastic if and only if the sum of the entries in each row is 1, i.e. $\sum_{i \in S} a_{ji} = 1$ for any $j \in S$.

EXERCISE 2.3. Show that if $P = [p_{ji}]_{j,i \in S}$ is a stochastic matrix, then any natural power P^n of P is a stochastic matrix. Is the corresponding result true for a double stochastic matrix?

REMARK 2.4. **Hint:** Show that if A and B are two stochastic matrices, then so is BA . For the second problem, recall that $(BA)^t = A^t B^t$.

EXERCISE 2.4. Let $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$. Show that

$$(16) \quad P^2 = \begin{bmatrix} 1+p^2-2p+pq & 2q-pq-q^2 \\ 2p-pq-p^2 & 1+q^2-2q+pq \end{bmatrix}.$$

REMARK 2.5. **Hint:** This is just simple matrix multiplication.

We see that there is a problem with finding higher powers of the matrix P . When multiplying P^2 by P , P^3 , and so on, we obtain more and more complicated expressions.

DEFINITION 2.6 (n -step transition matrix). The n -step transition matrix of a Markov chain ξ_n with transition probabilities $p(j \mid i)$, $j, i \in S$ is the matrix P_n with entries

$$(5.17) \quad p_n(j \mid i) = P(\xi_n = j \mid \xi_0 = i).$$

EXERCISE 2.5. Find an exact formula for P_n for the matrix P from Exercise ??.

REMARK 2.6. **Hint:** Put $x_n = P(\xi_n = 0 \mid \xi_0 = 0)$ and $y_n = P(\xi_n = 1 \mid \xi_0 = 1)$. Is it correct to suppose that $p_n(0 \mid 0) = x_n$ and $p_n(1 \mid 1) = y_n$? If yes, you may be able use Example ?? and Exercise ??.

EXERCISE 2.6. You may suspect that P_n equals P^n , the n th power of the matrix P . This holds for $n = 1$. Check if it is true for $n = 2$. If this is the case, try to prove that $P_n = P^n$ for all $n \in \mathbb{N}$.

REMARK 2.7. **Hint:** Once again, this is an exercise in matrix multiplication.

The following is a generalization of Exercise ??.

Proposition 2.7 (Chapman–Kolmogorov equation). *Suppose that $\xi_n, n \in \mathbb{N}$, is an S -valued Markov chain with n -step transition probabilities $p_n(j | i)$. Then for all $k, n \in \mathbb{N}$*

$$(5.18) \quad p_{n+k}(j | i) = \sum_{s \in S} p_n(j | s) p_k(s | i), \quad i, j \in S.$$

EXERCISE 2.7. *Prove Proposition ??.*

REMARK 2.8. **Hint:** $p_{n+k}(j | i)$ are the entries of the matrix $P_{n+k} = P^{n+k}$.

EXERCISE 2.8. *Suppose that $S = \mathbb{Z}$. Let $\eta_n, n \geq 1$ be a sequence of independent identically distributed random variables with $P(\eta_1 = 1) = p$ and $P(\eta_1 = -1) = q = 1 - p$. Define $\xi_n = \sum_{i=1}^n \eta_i$ for $n \geq 1$ and $\xi_0 = 0$. Show that ξ_n is a Markov chain with transition probabilities*

$$(17) \quad p(j | i) = \begin{cases} p, & \text{if } j = i + 1, \\ q, & \text{if } j = i - 1, \\ 0, & \text{otherwise.} \end{cases}$$

$\xi_n, n \geq 0$, is called a random walk starting at 0. Replacing $\xi_0 = 0$ with $\xi_0 = i$, we get a random walk starting at i .

REMARK 2.9. **Hint:** $\xi_{n+1} = \xi_n + \eta_{n+1}$. Are ξ_n and η_{n+1} independent?

EXERCISE 2.9. *For the random walk ξ_n defined in Exercise ?? prove that*

$$(5.19) \quad P(\xi_n = j | \xi_0 = i) = \binom{n}{\frac{n+j-i}{2}} p^{\frac{n+j-i}{2}} q^{\frac{n-j+i}{2}}$$

if $n + j - i$ is an even non-negative integer, and $P(\xi_n = j | \xi_0 = i) = 0$ otherwise.

REMARK 2.10. **Hint:** Use induction. Note that $\binom{n}{\frac{n+j-i}{2}} p^{\frac{n+j-i}{2}}$ equals 0 if $|j - i| \geq n + 1$.

Proposition 2.8. *For all $p \in (0, 1)$*

$$(5.20) \quad P(\xi_n = i | \xi_0 = i) \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Proposition 2.9. *The probability that the random walk ξ_n ever returns to the starting point is*

$$(18) \quad 1 - |p - q|.$$

EXERCISE 2.10. *Prove formula (5.24).*

REMARK 2.11. **Hint:** Use the Taylor formula to expand the right-hand side of (5.24) into a power series.

EXERCISE 2.11 (Branching process). *On the island Elschenbieden there lives an almost extinct species called Vugiel. Vugiel's males can produce zero, one, two, $\dots$ male offspring with probability $p_0, p_1, p_2, \dots$ respectively, where $p_i \geq 0$ and $\sum_{i=0}^{\infty} p_i = 1$. A challenging problem would be to find the Vugiel's chances of survival assuming that each individual lives exactly one year. At this moment, we ask you only to rewrite the problem in the language of Markov chains.*

REMARK 2.12. **Hint:** *The number of descendants of each male has the same distribution.*

EXERCISE 2.12. *Consider the following two cases:*

(1) *In Exercise ?? suppose that*

$$(19) \quad p_m = \binom{N}{m} p^m (1-p)^{N-m}.$$

for some $p \in (0, 1)$ and $N \in \mathbb{N}^$, where $\mathbb{N}^* = \{1, 2, 3, \dots\}$. (Note that $p_m = 0$ if $m > N$.) Show that*

$$(5.26) \quad p(j | i) = \binom{Ni}{j} p^j (1-p)^{Ni-j}.$$

Deduce that, in particular, $p(j | i) = 0$ if $j > Ni$.

(2) *Suppose that*

$$(20) \quad p_m = \frac{\lambda^m}{m!} e^{-\lambda}, m \in \mathbb{N},$$

for some $\lambda > 0$. In other words, assume that each X_j has the Poisson distribution with mean λ . Show that

$$(5.27) \quad p(j | i) = \frac{(\lambda i)^j}{j!} e^{-\lambda i}, j, i \geq 0.$$

REMARK 2.13. **Hint:** *If X_1 has the binomial distribution $P(X_1 = m) = \binom{N}{m} p^m (1-p)^{N-m}$, $m \in \mathbb{N}$, then there exists a finite sequence $\eta_1^1, \dots, \eta_N^1$ of independent identically distributed random variables such that $P(\eta_j^1 = 1) = p$, $P(\eta_j^1 = 0) = 1-p$ and $X_j = \eta_1^j + \dots + \eta_N^j = m$. Hence, $X_1 + \dots + X_i = \sum_{r=1}^i \sum_{s=1}^N \eta_s^r$, i.e. the sum of Ni independent identically distributed random variables with distribution as above. Hence we infer (5.26).*

Proposition 2.10. *The probability of survival in Exercise ??, part 2) equals 0 if $\lambda \leq 1$, and $1 - \hat{r}^k$ if $\lambda > 1$, where k is the initial Vugiel population and $\hat{r} \in (0, 1)$ is a solution to*

$$(5.28) \quad r = e^{(r-1)\lambda}.$$

REMARK 2.14. *The method presented in the last solution works for any distribution of the variables X_j . It turns out that the mean value λ of X_1 plays the same role as above. One can show that if $\lambda \leq 1$, then the population will become extinct with probability 1, while for $\lambda > 1$ the probability of extinction is larger than 0 and smaller than 1.*

EXERCISE 2.13. On the, now familiar, island of Elschénbieden the question of survival of the Vugiel is a hot political issue. The (human) population of the island is N . Those who believe that action should be taken in order to help the animals preach their conviction quite convincingly. For if a supporter discusses the issue with a non-supporter, the latter will change his mind with probability one. However, they do so only in face-to-face encounters. Suppose that the probability of an encounter of exactly one pair of humans during one day is p and that with probability q this pair is a supporter-non-supporter one. Write down a Markov chain model of this situation. Neglect the probability of two or more encounters during one day.

REMARK 2.15. **Hint:** On each day the number of supporters can either increase by 1 or remain unchanged. What is the probability of the former?

EXERCISE 2.14 (Queuing model). A car wash machine can serve at most one customer at a time. With probability p , $0 < p < 1$, the machine can finish serving a customer in a unit time. If this happens, the next waiting car (if any) can be served at the beginning of the next unit of time. During the time interval between the n th and $(n+1)$ st unit of time the number of cars arriving has the Poisson distribution with parameter $\lambda > 0$. Let ξ_n denote the number of cars being served or waiting to be served at the beginning of unit n . Show that $\xi_n, n \in \mathbb{N}$, is a Markov chain and find its transition probabilities.

REMARK 2.16. **Hint:** Let $Z_n, n = 0, 1, 2, \dots$ be a sequence of independent identically distributed random variables, each having the Poisson distribution with parameter λ . Then $\xi_{n+1} - \xi_n - Z_n$ equals -1 or 0 .

REMARK 2.17. In the last model we are interested in the behaviour of ξ_n for large values of n . In particular, it is interesting to determine whether the limit of ξ_n or that of $\mathbb{E}\xi_n$ (as $n \rightarrow \infty$) exists. In Exercise ?? we shall find conditions which guarantee the existence of a unique invariant measure and imply that the Markov chain in question is ergodic.

2. Classification of States

In what follows we fix an S -valued Markov chain with transition matrix $P = [p(j | i)]_{j,i \in S}$, where S is a non-empty and at most countable set.

DEFINITION 2.11 (Recurrent and transient states). A state i is called recurrent if the process ξ_n will eventually return to i given that it starts at i , i.e.

$$(5.36) \quad P(\xi_n = i \text{ for some } n \geq 1 \mid \xi_0 = i) = 1.$$

If the condition (5.36) is not satisfied, then the state i is called transient.

Theorem 2.12. Show that for a random walk on $\mathbb{Z}$ with parameter $p \in (0, 1)$, the state 0 is recurrent if and only if $p = 1/2$. Show that the same holds if 0 is replaced by any other state $i \in \mathbb{Z}$.

DEFINITION 2.13 (Communicating states). We say that a state i communicates with a state j if with positive probability the chain will visit the state j having started at i , i.e.

$$(5.37) \quad P(\xi_n = j \text{ for some } n \geq 0 \mid \xi_0 = i) > 0.$$

If i communicates with j , then we shall write $i \rightarrow j$. We say that the state i intercommunicates with a state j , and write $i \leftrightarrow j$, if $i \rightarrow j$ and $j \rightarrow i$.

EXERCISE 2.15. Show that $i \rightarrow j$ if and only if $p_k(j | i) > 0$ for some $k \geq 1$.

REMARK 2.18. **Hint:** Recall that $p_k(j | i) = P(\xi_k = j | \xi_0 = i)$.

EXERCISE 2.16. Show that

- (1) $i \leftrightarrow i$,
- (2) if $i \leftrightarrow j$ then $j \leftrightarrow i$, and
- (3) if $i \leftrightarrow j, j \leftrightarrow k$ then $i \leftrightarrow k$.

In other words, show that $\leftrightarrow$ is an equivalence relation on S .

REMARK 2.19. **Hint:** 1) and 2) are obvious. For 3) use the Chapman-Kolmogorov equations.

EXERCISE 2.17. For $|x| < 1$ and $j, i \in S$ define

$$(5.38) \quad P_{ji}(x) = \sum_{n=0}^{\infty} p_n(j | i) x^n,$$

$$(5.39) \quad F_{ji}(x) = \sum_{n=1}^{\infty} f_n(j | i) x^n,$$

where $f_n(j | i) = P(\xi_n = j, \xi_k \neq j, k = 1, \dots, n-1 | \xi_0 = i)$. Show that the power series in (5.38)-(5.39) are absolutely convergent for $|x| < 1$ and that

$$(5.40) \quad P_{ji}(x) = F_{ji}(x)P_{jj}(x), \text{ if } j \neq i,$$

$$(5.41) \quad P_{ii}(x) = 1 + F_{ii}(x)P_{ii}(x).$$

REMARK 2.20. **Hint:** Note that $|p_n(j | i)| \leq 1$, so the radius of convergence of the power series (5.38) is ≥ 1 .

EXERCISE 2.18. Show that $\lim_{x \rightarrow 1} P_{jj}(x) = \sum_{n=0}^{\infty} p_n(j | j)$ and $\lim_{x \rightarrow 1} F_{jj}(x) = \sum_{n=0}^{\infty} f_n(j | j)$.

REMARK 2.21. **Hint:** Apply Abel's lemma: If $a_k \geq 0$ for all $k \geq 0$ and $\limsup_{k \rightarrow \infty} \sqrt[k]{a_k} \leq 1$, then $\lim_{x \rightarrow 1} \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k$, no matter whether this sum is finite or infinite.

EXERCISE 2.19. Show that a state j is recurrent if and only if $\sum_n p_n(j | j) = \infty$. Deduce that the state j is transient if and only if

$$(5.42) \quad \sum_n p_n(j | j) < \infty.$$

Show that if j is transient, then for each $i \in S$

$$(5.43) \quad \sum_n p_n(j | i) < \infty.$$

REMARK 2.22. **Hint:** If j is recurrent, then $F_{jj}(x) \rightarrow \sum_n f_n(j | j) = 1$ as $x \nearrow 1$. Use (5.41) in conjunction with Abel's Lemma.

EXERCISE 2.20. For a Markov chain ξ_n with transition matrix $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$ show that both states are recurrent.

REMARK 2.23. **Hint:** Use Exercise ?? and Exercise ??.

One may suspect that if the state space S is finite, then there must exist at least one recurrent state. For otherwise, if all states were transient and $S = \{1, 2, \dots, N\}$, then with positive probability a chain starting from 1 would visit 1 only a finite number of times. Thus, after visiting that state for the last time, the chain would move to a different state, say i_2 , in which it would stay for a finite time only with positive probability. Thus, in finite time, with positive probability, the chain will never return to states 1 and i_2 . By induction, in finite time, with positive probability, the chain will never return to any of the states. This is impossible. The following exercise will give precision to this argument.

EXERCISE 2.21. Show that if ξ_n is a Markov chain with finite state space S , then there exists at least one recurrent state $i \in S$.

REMARK 2.24. **Hint:** Argue by contradiction and use (5.43).

The following result is quoted here for reference. The proof is surprisingly difficult and falls beyond the scope of this book.

Theorem 2.14. A state $j \in S$ is recurrent if and only if

$$(21) \quad P(\xi_n = j \text{ for infinitely many } n \mid \xi_0 = j) = 1,$$

and is transient if and only if

$$(22) \quad P(\xi_n = j \text{ for infinitely many } n \mid \xi_0 = j) = 0.$$

DEFINITION 2.15 (Null-recurrent and positive-recurrent states). For an S -valued Markov chain $\xi_n, n \in \mathbb{N}$, a state $i \in S$ is called null-recurrent if it is recurrent and its mean recurrence time m_i defined by

$$(5.44) \quad m_i := \sum_{n=0}^{\infty} n f_n(i | i)$$

equals ∞ . A state $i \in S$ is called positive-recurrent if it is recurrent and its mean recurrence time m_i is finite.

REMARK 2.25. One can show that a recurrent state i is null-recurrent if and only if $p_n(i | i) \rightarrow 0$.

We already know that for a random walk on $\mathbb{Z}$ the state 0 is recurrent if and only if $p = 1/2$, i.e. if and only if the random walk is symmetric. In the following problem we shall try to answer if 0 is a null-recurrent or positive-recurrent state (when $p = 1/2$).

EXERCISE 2.22. Consider a symmetric random walk on $\mathbb{Z}$. Show that 0 is a null-recurrent state. Can you deduce whether other states are positive-recurrent or null-recurrent?

REMARK 2.26. **Hint:** State 0 is null-recurrent if and only if $\sum_n n f_n(0 | 0) = \infty$. As in Exercise ??, $\sum_n n f_n(0 | 0) = \lim_{x \rightarrow 1} F'_{00}(x)$, where F_{00} is defined by (5.39).

EXERCISE 2.23. For the Markov chain ξ_n from Exercise ?? show that not only are all states recurrent, but they are positive-recurrent.

REMARK 2.27. **Hint:** Calculate $f_n(0 | 0)$ directly.

The last two exercises suggest that the type of a state $i \in S$, i.e. whether it is transient, null-recurrent or positive-recurrent is invariant under the equivalence $\leftrightarrow$. We shall investigate this question in more detail below, but even before doing so we need one more notion: that of a periodic state.

DEFINITION 2.16 (Periodic and aperiodic states). Suppose that $\xi_n, n \in \mathbb{N}$, is a Markov chain on a state space S . Let $i \in S$. We say that i is a periodic state if and only if the greatest common divisor (gcd) of all $n \in \mathbb{N}^*$, where $\mathbb{N}^* = \{1, 2, 3, \dots\}$, such that $p_n(i | i) > 0$ is ≥ 2 . Otherwise, the state i is called aperiodic. In both cases, the gcd is denoted by $d(i)$ and is called the period of the state i . Thus, i is periodic if and only if $d(i) \geq 2$. A state i which is positive recurrent and aperiodic is called ergodic.

EXERCISE 2.24. Is this claim that $p_{d(i)}(i | i) > 0$ true or not?

REMARK 2.28. **Hint:** Think of a Markov chain in which it is possible to return to the starting point by two different routes. One route with four steps, the other one with six steps.

One of the by-products of the following exercise is another example of the type asked for in Exercise ??.

EXERCISE 2.25. Consider a Markov chain on $S = \{1, 2\}$ with transition probability matrix $P = \begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix}$. This chain can also be described by the graph in Figure ??. Find $d(1)$ and $d(2)$.

REMARK 2.29. **Hint:** Calculate P^2 and P^3 . This can be done in two different ways: either algebraically or, probabilistically.

Proposition 2.17. Suppose that $i, j \in S$ and $i \leftrightarrow j$. Show that

- (1) i is transient if and only if j is;
- (2) i is recurrent if and only if j is;
- (3) i is null-recurrent if and only if j is;
- (4) i is positive-recurrent if and only if j is;
- (5) i is periodic if and only if j is, in which case $d(i) = d(j)$;
- (6) i is ergodic if and only if j is.

EXERCISE 2.26. Show that the following modification of 2) above is true: If i is recurrent and $i \rightarrow j$, then $j \rightarrow i$. Deduce that if i is recurrent and $i \rightarrow j$, then j is recurrent and $j \leftrightarrow i$.

REMARK 2.30. **Hint:** Is it possible for a chain starting from i to visit j and then never return to i ? Is such a situation possible when i is a recurrent state?

The following result describes how the state space S can be partitioned into a countable sum of classes. One of these classes consists of all transient states. Each of the other classes

consists of interconnecting recurrent states. If the chain enters one of the classes of second type, it will never leave it. However, if the chain enters the class of transient states, it will eventually leave it (and so never return to it). We begin with a definition.

DEFINITION 2.18 (Closed and irreducible sets). *Suppose that $\xi_n, n \in \mathbb{N}$, is a Markov chain on a countable state space S .*

(1) *A set $C \subset S$ is called closed if once the chain enters C it will never leave it, i.e.*

$$(5.45) \quad P(\xi_k \in S \setminus C \text{ for some } k \geq n \mid \xi_n \in C) = 0.$$

(2) *A set $C \subset S$ is called irreducible if any two elements i, j of C intercommunicate, i.e. for all $i, j \in C$ there exists an $n \in \mathbb{N}$ such that $p_n(j \mid i) > 0$.*

Theorem 2.19. *Suppose that $\xi_n, n \in \mathbb{N}$, is a Markov chain on a countable state space S . Then*

$$(5.46) \quad S = T \cup \bigcup_{j=1}^N C_j, \text{ (disjoint sum),}$$

where T is the set of all transient states in S and each C_j is a closed irreducible set of recurrent states.

EXERCISE 2.27. *Suppose that $\xi_n, n \in \mathbb{N}$, is a Markov chain on a countable state space S . Show that a set $C \subset S$ is closed if and only if $p(j \mid i) = 0$ for all $i \in C$ and $j \in S \setminus C$.*

REMARK 2.31. *Hint:* One implication is trivial. For the other one use the countable additivity of the measure P .

3. Long-Time Behaviour of Markov Chains: General Case

For convenience we shall denote the countable state space S by $\{1, 2, 3, \dots\}$ when S is an infinite set and by $\{1, 2, \dots, n\}$ when S is finite.

Proposition 2.20. *Let $P = [p(j \mid i)]$ be the transition matrix of a Markov chain with state space S . Suppose that for all $i, j \in S$*

$$(5.47) \quad \lim_{n \rightarrow \infty} p_n(j \mid i) =: \pi_j.$$

(In particular, the limit is independent of i .) Then

- (1) $\sum_j \pi_j \leq 1$;
- (2) $\sum_i p(j \mid i) \pi_i = \pi_j$;
- (3) either $\sum_j \pi_j = 1$, or $\pi_j = 0$ for all $j \in S$.

Proposition 2.21. *Let $P = [p(j \mid i)]$ be the transition matrix of a Markov chain with state space S . Suppose that for all $i, j \in S$*

$$(5.47) \quad \lim_{n \rightarrow \infty} p_n(j \mid i) =: \pi_j.$$

Then the vector $\pi = (\pi_j)_{j \in S}$ satisfies $\pi = P^T \pi$, i.e.,

$$(5.65) \quad \sum_{i \in S} p(j | i) \pi_i = \pi_j \quad \text{for all } j \in S.$$

Theorem 2.22. Suppose that $a \geq 1$ is a natural number. Consider a random walk on $S = \{0, 1, \dots, a\}$ with absorbing barriers at 0 and a , and with probability p of moving to the right and probability $q = 1 - p$ of moving to the left from any of the states $1, \dots, a - 1$. Hence, our random walk is a Markov chain with transition probabilities

$$(23) \quad p(j | i) = \begin{cases} p, & \text{if } 1 \leq i \leq a - 1, j = i + 1, \\ q, & \text{if } 1 \leq i \leq a - 1, j = i - 1, \\ 1, & \text{if } i = j = 0 \text{ or } i = j = a, \\ 0, & \text{otherwise.} \end{cases}$$

Find, (a) all invariant measures (there may be just one), (b) the probability of hitting the right-hand barrier prior to hitting the left-hand one.

REMARK 2.32. **Hint:** For (a) recall Exercise ?? and for (b) Exercise ??.

EXERCISE 2.28. Ian plays a fair game of dice. After the n th roll of a die he writes down the maximum outcome ξ_n obtained so far. Show that ξ_n is a Markov chain and find its transition probabilities.

REMARK 2.33. **Hint:** $\xi_{n+1} = \max\{\xi_n, X_{n+1}\}$, where X_k is the outcome of the k th roll.

EXERCISE 2.29. Analyse the Markov chain described in Exercise ??, but with fair die replaced by a fair pyramid.

REMARK 2.34. **Hint:** A pyramid has four faces only.

EXERCISE 2.30. Consider a random walk on $S = \{0, 1, \dots, a\}$ with absorbing barriers at 0 and a , and with probability p of moving to the right and probability $q = 1 - p$ of moving to the left from any of the states $1, \dots, a - 1$. Find the probability of hitting the right-hand barrier before the left-hand one.

REMARK 2.35. **Hint:** Use the result from Proposition ?? or solve directly using recurrence relations.

Proposition 2.23. For the random walk with absorbing barriers described above, the probability of hitting state a before state 0 when starting from i ($1 \leq i \leq a - 1$) is given by

$$(24) \quad P(\text{hit } a \text{ before } 0 \mid \xi_0 = i) = \frac{1 - (q/p)^i}{1 - (q/p)^a} \quad \text{if } p \neq q,$$

and

$$(25) \quad P(\text{hit } a \text{ before } 0 \mid \xi_0 = i) = \frac{i}{a} \quad \text{if } p = q = 1/2.$$

Proposition 2.24. The stationary distribution π for the random walk with absorbing barriers (when considered on the non-absorbing states) satisfies

$$(26) \quad \pi_i = \frac{1}{Z} \left(\frac{p}{q} \right)^i, \quad i = 0, 1, \dots, a,$$

where $Z = \sum_{i=0}^a (p/q)^i$ is the normalizing constant.

Theorem 2.25. *For the Poisson queueing model in Exercise ??, the stationary distribution exists if and only if $\lambda' < 1$, where λ' is the mean arrival rate. When it exists, the stationary distribution is Poisson with parameter λ' , i.e.,*

$$(27) \quad \pi_j = e^{-\lambda'} \frac{(\lambda')^j}{j!}, \quad j = 0, 1, 2, \dots$$

EXERCISE 2.31. *Prove that if there exists an invariant measure for the Markov chain in Exercise ??, then $\lambda' := \sum_{j=0}^{\infty} j q'_j < 1$. Assuming that the converse is also true, conclude that the chain is ergodic if and only if $\lambda' < 1$. Show that if such an invariant measure exists, then it is unique.*

REMARK 2.36. **Hint:** Suppose that $\pi = \sum_{j=0}^{\infty} \pi_j \delta_j$ is an invariant measure. Write down an infinite system of linear equations for π_j . If you don't know how to follow, look at the solution.

4. Long-Time Behaviour of Markov Chains with Finite State Space

As we have seen above, the existence of the π_j plays a very important role in the study of invariant measures. In what follows we shall investigate this question in the case when the state space S is finite.

Theorem 2.26. *Suppose that S is finite and the transition matrix $P = [p(j | i)]$ of a Markov chain on S satisfies the condition*

$$(5.54) \quad \exists n_0 \in \mathbb{N} \exists \varepsilon > 0 : p_{n_0}(j | i) \geq \varepsilon, \quad i, j \in S.$$

Then, the following limit exists for all $i, j \in S$ and is independent of i :

$$(5.55) \quad \lim_{n \rightarrow \infty} p_n(j | i) = \pi_j.$$

The numbers π_j satisfy

$$(5.56) \quad \pi_j > 0, j \in S \text{ and } \sum_{j \in S} \pi_j = 1.$$

Conversely, if a sequence of numbers $\pi_j, j \in S$ satisfies conditions (5.55)-(5.56), then assumption (5.54) is also satisfied.

EXERCISE 2.32. *Show that $p_n(j | i) \rightarrow \pi_j$ at an exponential rate.*

REMARK 2.37. **Hint:** Recall that $m_n(j) \leq \pi_j \leq M_n(j)$ and use (5.62).

The above proves the following important result.

Theorem 2.27. Suppose that the transition matrix $P = [p(j | i)]$ of a Markov chain $\xi_n, n \in \mathbb{N}$, satisfies assumption (5.54). Show that there exists a unique invariant measure μ . Moreover, for some $A > 0$, and $\alpha < 1$

$$(5.63) \quad |p_n(j | i) - \pi_j| \leq A\alpha^n, \quad i, j \in S, n \in \mathbb{N}.$$

EXERCISE 2.33. Investigate the existence and uniqueness of an invariant measure for the Markov chain in Proposition ??.

REMARK 2.38. **Hint:** Are the assumptions of Theorem ?? satisfied?

REMARK 2.39. The solution to Exercise ?? allows us to find the unique invariant measure by direct methods, i.e. by solving the linear equations (5.79)-(5.80).

EXERCISE 2.34. Find the invariant measure from Exercise ?? by calculating the limits (5.55).

REMARK 2.40. **Hint:** Refer to Solution 5.5.

EXERCISE 2.35. For each of the following transition matrices, determine: (a) whether the chain is irreducible, (b) whether it is aperiodic, (c) the invariant distribution (if it exists), (d) whether the chain is ergodic.

$$\begin{aligned} (1) \quad P &= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \\ (2) \quad P &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ (3) \quad P &= \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

REMARK 2.41. **Hint:** Check the definitions of irreducibility, aperiodicity, and use Theorem ??.

CHAPTER 3

Stochastic Processes in Continuous Time

1. General Notions

The following definitions are straightforward extensions of those introduced earlier for sequences of random variables, the underlying idea being that of a family of random variables depending on time.

DEFINITION 3.1 (Stochastic Process). *A stochastic process is a family of random variables $\xi(t)$ parametrized by $t \in T$, where $T \subset \mathbb{R}$. When $T = \{1, 2, \dots\}$, we shall say that $\xi(t)$ is a stochastic process in discrete time (i.e. a sequence of random variables). When T is an interval in $\mathbb{R}$ (typically $T = [0, \infty)$), we shall say that $\xi(t)$ is a stochastic process in continuous time.*

For every $\omega \in \Omega$ the function

$$T \ni t \mapsto \xi(t, \omega)$$

is called a path (or sample path) of $\xi(t)$.

DEFINITION 3.2 (Filtration). *A family $\mathcal{F}_t$ of σ -fields on Ω parametrized by $t \in T$, where $T \subset \mathbb{R}$, is called a filtration if*

$$\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$$

for any $s, t \in T$ such that $s \leq t$.

DEFINITION 3.3 (Martingale). *A stochastic process $\xi(t)$ parametrized by $t \in T$ is called a martingale (submartingale, supermartingale) with respect to a filtration $\mathcal{F}_t$ if*

- (1) $\xi(t)$ is integrable for each $t \in T$;
- (2) $\xi(t)$ is $\mathcal{F}_t$ -measurable for each $t \in T$ (in which case we say that $\xi(t)$ is adapted to $\mathcal{F}_t$);
- (3) $\xi(s) = E(\xi(t) \mid \mathcal{F}_s)$ (respectively, $\leq$ or $\geq$) for every $s, t \in T$ such that $s \leq t$.

In earlier chapters we have seen various stochastic processes, in particular, martingales in discrete time such as the symmetric random walk, for example. In what follows we shall study in some detail two processes in continuous time, namely, the Poisson process and Brownian motion.

2. Poisson Process

2.1. Exponential Distribution and Lack of Memory.

DEFINITION 3.4 (Exponential Distribution). *We say that a random variable η has the exponential distribution of rate $\lambda > 0$ if*

$$(28) \quad P\{\eta > t\} = e^{-\lambda t}$$

for all $t \geq 0$.

For example, the emissions of particles by a sample of radioactive material (or calls made at a telephone exchange) occur at random times. The probability that no particle is emitted (no call is made) up to time t is known to decay exponentially as t increases. That is to say, the time η of the first emission has the exponential distribution, $P\{\eta > t\} = e^{-\lambda t}$.

EXERCISE 3.1. *What is the distribution function of a random variable η with exponential distribution? Does it have a density? If so, find the density.*

Hint: *What is the probability that $\eta > 0$? What is the probability that $\eta > t$ for any given $t < 0$? Can you express the distribution function in terms of $P\{\eta > t\}$? Is the distribution function differentiable?*

EXERCISE 3.2. *Compute the expectation and variance of a random variable having the exponential distribution.*

Hint: *Use the density found in Exercise ??.*

EXERCISE 3.3. *Show that a random variable η with exponential distribution satisfies*

$$(6.1) \quad P\{\eta > t + s\} = P\{\eta > t\}P\{\eta > s\}$$

for any $s, t \geq 0$.

Hint: *When the probabilities are replaced by exponents, the equality should become obvious.*

EXERCISE 3.4. *Show that the equality in Exercise ?? is equivalent to*

$$(6.2) \quad P\{\eta > t + s \mid \eta > s\} = P\{\eta > t\}$$

for any $s, t \geq 0$.

Hint: *Recall how to compute conditional probability. Observe that $\eta > s + t$ implies $\eta > s$.*

The equality ?? (or, equivalently, ??) is known as the **lack of memory property**. The odds that no particle will be emitted (no call will be made) in the next time interval of length t are not affected by the length of time s it has already taken to wait, given that no emission (no call) has occurred yet.

EXERCISE 3.5. *Show that the exponential distribution is the only probability distribution satisfying the lack of memory property.*

Hint: *The lack of memory property means that $g(t) = P\{\eta > t\}$ satisfies the functional equation*

$$g(t + s) = g(t)g(s)$$

for any $s, t > 0$. Find all non-negative non-increasing solutions of this functional equation.

2.2. Construction of the Poisson Process. Let $\eta_1, \eta_2, \dots$ be a sequence of independent random variables, each having the same exponential distribution of rate λ . For example, the times between the emissions of radioactive particles (or between calls made at a telephone exchange) have this property. We put

$$\xi_n = \eta_1 + \dots + \eta_n,$$

which can be thought of as the time of the n -th emission (the n -th call). We also put $\xi_0 = 0$ for convenience. The number of emissions (calls) up to time $t \geq 0$ is an n such that $\xi_{n+1} > t \geq \xi_n$. In other words, the number of emissions (calls) up to time $t \geq 0$ is equal to $\max\{n : t \geq \xi_n\}$.

DEFINITION 3.5 (Poisson Process). *We say that $N(t)$, where $t \geq 0$, is a Poisson process if*

$$(29) \quad N(t) = \max\{n : t \geq \xi_n\}.$$

Thus, $N(t)$ can be regarded as the number of particles emitted (calls made) up to time t . It is an example of a stochastic process in continuous time. A typical path of $N(t)$ is shown in ???. It begins at the origin, $N(0) = 0$ (no particles emitted at time 0), and is right-continuous, non-decreasing and piecewise constant with jumps of size 1 at the times ξ_n .

FIGURE 1. A typical path of $N(t)$ and the jump times ξ_n

What is the distribution of $N(t)$? To answer this question we need to recall the definition of the Poisson distribution.

DEFINITION 3.6 (Poisson Distribution). *A random variable ν has the Poisson distribution with parameter $\alpha > 0$ if*

$$(30) \quad P\{\nu = n\} = e^{-\alpha} \frac{\alpha^n}{n!}$$

for any $n = 0, 1, 2, \dots$

The probabilities $P\{\nu = n\}$ for various values of α are shown in ???.

FIGURE 2. Poisson distribution with parameter α

Proposition 3.7 (Poisson Process Distribution). *$N(t)$ has the Poisson distribution with parameter λt ,*

$$(31) \quad P\{N(t) = n\} = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

PROOF. *First of all, observe that*

$$(32) \quad \{N(t) < n\} = \{\xi_n > t\}.$$

It suffices to compute the probability of this event for any n because

$$(6.3) \quad \begin{aligned} P\{N(t) = n\} &= P\{N(t) < n+1\} - P\{N(t) < n\} \\ &= P\{\xi_{n+1} > t\} - P\{\xi_n > t\}. \end{aligned}$$

We shall prove by induction on n that

$$(6.4) \quad P\{\xi_n > t\} = e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}.$$

For $n = 1$:

$$P\{\xi_1 > t\} = P\{\eta_1 > t\} = e^{-\lambda t}.$$

Next, suppose that ?? holds for some n . Then, expressing ξ_{n+1} as the sum of the independent random variables ξ_n and η_{n+1} , we compute

$$\begin{aligned}
 P\{\xi_{n+1} > t\} &= P\{\xi_n + \eta_{n+1} > t\} \\
 &= P\{\eta_{n+1} > t\} + P\{\xi_n > t - \eta_{n+1}, t \geq \eta_{n+1} > 0\} \\
 &= e^{-\lambda t} + \int_0^t P\{\xi_n > t - s\} f_{\eta_{n+1}}(s) ds \\
 &= e^{-\lambda t} + \int_0^t e^{-\lambda(t-s)} \sum_{k=0}^{n-1} \frac{(\lambda(t-s))^k}{k!} \lambda e^{-\lambda s} ds \\
 &= e^{-\lambda t} + e^{-\lambda t} \sum_{k=0}^{n-1} \frac{\lambda^{k+1}}{k!} \int_0^t (t-s)^k ds \\
 &= e^{-\lambda t} \sum_{k=0}^n \frac{(\lambda t)^k}{k!},
 \end{aligned}
 \tag{33}$$

where $f_{\eta_{n+1}}(s)$ is the density of η_{n+1} . By induction, ?? holds for any n . Now apply ?? to complete the proof.

EXERCISE 3.6. What is the expectation of $N(t)$?

Hint: What are the possible values of $N(t)$? What are the corresponding probabilities? Can you compute the expectation from these? To simplify the result use the Taylor expansion of e^x .

EXERCISE 3.7. Compute $P\{N(s) = 1, N(t) = 2\}$ for any $0 \leq s < t$.

Hint: Express $\{N(s) = 1, N(t) = 2\}$ as $\{\eta_1 \leq s < \eta_1 + \eta_2 \leq t < \eta_1 + \eta_2 + \eta_3\}$. You can compute the probability of the latter, since η_1, η_2, η_3 are exponentially distributed and independent.

2.3. Poisson Process Starts from Scratch at Time t . Imagine that you are to take part in an experiment to count the emissions of a radioactive particle. Unfortunately, in the excitement of proving the lack of memory property you forget about the engagement and arrive late to find that the experiment has already been running for time t and you have missed the first $N(t)$ emissions. Determined to make the best of it, you start counting right away, so at time $t + s$ you will have registered $N(t + s) - N(t)$ emissions. It will now be necessary to discuss $N(t + s) - N(t)$ instead of $N(s)$ in your report.

What are the properties of $N(t + s) - N(t)$? Perhaps you can guess something from the physical picture? After all, a sample of radioactive material will keep emitting particles no matter whether anyone cares to count them or not. So the moment when someone starts counting does not seem important. You can expect $N(t + s) - N(t)$ to behave in a similar way as $N(s)$. And because radioactive emissions have no memory of the past, $N(t + s) - N(t)$ should be independent of $N(t)$.

To study this conjecture recall the construction of a Poisson process $N(t)$ based on a sequence of independent random variables $\eta_1, \eta_2, \dots$, all having the same exponential distribution. We shall try to represent $N(t + s) - N(t)$ in a similar way.

Let us put

$$\eta_1^t := \xi_{N(t)+1} - t, \quad \eta_n^t := \eta_{N(t)+n}, \quad n = 2, 3, \dots,
 \tag{34}$$

see ???. These are the times between the jumps of $N(t+s) - N(t)$. Then we define

$$(35) \quad \xi_n^t = \eta_1^t + \cdots + \eta_n^t,$$

$$(36) \quad N^t(s) = \max\{n : \xi_n^t \leq s\}.$$

FIGURE 3. The random variables $\eta_1^t, \eta_2^t, \eta_3^t, \dots$

EXERCISE 3.8. Show that

$$(37) \quad N^t(s) = N(t+s) - N(t).$$

Hint: First show that $\xi_n^t = \xi_{N(t)+n} - t$.

If we can show that $\eta_1^t, \eta_2^t, \dots$ are independent random variables having the same exponential distribution as $\eta_1, \eta_2, \dots$, it will mean that $N(t+s) - N(t)$ is a Poisson process with the same probability distribution as $N(s)$. Moreover, if the η_n^t turn out to be independent of $N(t)$, it will imply that $N(t+s) - N(t)$ is also independent of $N(t)$.

Before setting about this task beware of one common mistake. It is sometimes claimed that the times $\eta_1^t, \eta_2^t, \eta_3^t, \dots$ between the jumps of $N(t+s) - N(t)$ are equal to $\xi_{n+1} - t, \eta_{n+2}, \eta_{n+3}, \dots$ for some n . Hence $\eta_1^t, \eta_2^t, \eta_3^t, \dots$ are independent because the random variables $\xi_{n+1}, \eta_{n+2}, \eta_{n+3}, \dots$ are. The flaw in this is that, in fact, $\eta_1^t, \eta_2^t, \eta_3^t, \dots$ are equal to $\xi_{n+1} - t, \eta_{n+2}, \eta_{n+3}, \dots$ only on the set $\{N(t) = n\}$. However, the argument can be saved by conditioning with respect to $N(t)$. Our task becomes an exercise in computing conditional probabilities.

EXERCISE 3.9. Show that

$$(38) \quad P\{\eta_1^t > s \mid N(t)\} = P\{\eta_1 > s\}.$$

Hint: It suffices (why?) to verify that

$$P\{\eta_1^t > s, N(t) = n\} = P\{\eta_1 > s\}P\{N(t) = n\}$$

for any n . To this end, write the sets $\{N(t) = n\}$ and $\{\eta_1^t > s, N(t) = n\}$ in terms of ξ_n and η_{n+1} , which are independent random variables, and use the lack of memory property for η_{n+1} .

EXERCISE 3.10. Show that

$$(39) \quad P\{\eta_1^t > s_1, \dots, \eta_k^t > s_k \mid N(t)\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\}.$$

Hint: Verify that

$$P\{\eta_1^t > s_1, \eta_2^t > s_2, \dots, \eta_k^t > s_k, N(t) = n\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\}P\{N(t) = n\}$$

for any n . This is done in Exercise ?? for $k = 1$.

EXERCISE 3.11. From the formula in Exercise ?? deduce that the random variables η_n^t and $N(t)$ are independent, and the η_n^t have the same probability distribution as η_n , i.e. exponential with parameter λ .

Hint: What do you get if you take the expectation on both sides of the equality in Exercise ??? Can you deduce the probability distribution of η_n^t ? Can you see that the η_n^t are independent?

To prove that the η_n^t are independent of $N(t)$ you need to be a little more careful and integrate over $\{N(t) = n\}$ instead of taking the expectation.

Because $N(t+s) - N(t)$ can be defined in terms of $\eta_1^t, \eta_2^t, \dots$ in the same way the original Poisson process $N(t)$ is defined in terms of $\eta_1, \eta_2, \dots$, the result in Exercise ?? proves the theorem below.

Theorem 3.8 (Poisson Process Stationarity). *For any fixed $t \geq 0$*

$$N^t(s) = N(t+s) - N(t), \quad s \geq 0$$

is a Poisson process independent of $N(t)$ with the same probability law as $N(s)$.

That is to say, for any $s, t \geq 0$ the increment $N(t+s) - N(t)$ is independent of $N(t)$ and has the same probability distribution as $N(s)$. The assertion can be generalized to several increments, resulting in the following important theorem.

Theorem 3.9 (Poisson Process Independent Increments). *For any $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$ the increments*

$$N(t_1), N(t_2) - N(t_1), N(t_3) - N(t_2), \dots, N(t_n) - N(t_{n-1})$$

are independent and have the same probability distribution as

$$N(t_1), N(t_2 - t_1), N(t_3 - t_2), \dots, N(t_n - t_{n-1}).$$

PROOF. From ?? it follows immediately that each increment $N(t_i) - N(t_{i-1})$ has the same distribution as $N(t_i - t_{i-1})$ for $i = 1, \dots, n$.

It remains to prove independence. This can be done by induction. The case when $n = 2$ is covered by ??. Now suppose that independence holds for n increments of a Poisson process for some $n \geq 2$. Take any sequence

$$0 \leq t_1 \leq t_2 \leq \dots \leq t_n \leq t_{n+1}.$$

By the induction hypothesis

$$N(t_{n+1}) - N(t_n), \dots, N(t_2) - N(t_1)$$

are independent, since they can be regarded as increments of

$$N^{t_1}(s) = N(t_1 + s) - N(t_1),$$

which is a Poisson process by ??. By the same theorem these increments are independent of $N(t_1)$. It follows that the $n+1$ random variables

$$N(t_{n+1}) - N(t_n), \dots, N(t_2) - N(t_1), N(t_1)$$

are independent, completing the proof.

DEFINITION 3.10 (Independent Increments). *We say that a stochastic process $\xi(t)$, where $t \in T$, has independent increments if*

$$\xi(t_1) - \xi(t_0), \dots, \xi(t_n) - \xi(t_{n-1})$$

are independent for any $t_0 < t_1 < \dots < t_n$ such that $t_0, t_1, \dots, t_n \in T$.

DEFINITION 3.11 (Stationary Increments). *A stochastic process $\xi(t)$, where $t \in T$, is said to have stationary increments if for any $s, t \in T$ the probability distribution of $\xi(t+h) - \xi(s+h)$ is the same for each h such that $s+h, t+h \in T$.*

?? implies that the Poisson process has stationary independent increments. The result in the next exercise is also a consequence of ??.

EXERCISE 3.12. Show that $N(t) - \lambda t$ is a martingale with respect to the filtration $\mathcal{F}_t$ generated by the family of random variables $\{N(s) : s \in [0, t]\}$.

Hint: Observe that $N(t) - N(s)$ is independent of $\mathcal{F}_s$ by ??.

2.4. Various Exercises on the Poisson Process.

EXERCISE 3.13. Show that $\xi_0 < \xi_1 < \xi_2 < \dots$ a.s.

Hint: What is the probability of the event $\{\xi_{n-1} < \xi_n\} = \{\eta_n > 0\}$? What is the probability of the intersection of all such events?

EXERCISE 3.14. Show that $\lim_{n \rightarrow \infty} \xi_n = \infty$ a.s.

Hint: If $\lim_{n \rightarrow \infty} \xi_n < \infty$, then the sequence $\eta_1, \eta_2, \dots$ of independent random variables, all having the same exponential distribution, must be bounded. What is the probability that such a sequence is bounded? Begin with computing the probability $P\{\eta_1 \leq m, \dots, \eta_n \leq m\}$ for any fixed $m > 0$.

Although it is instructive to estimate $P\{\lim_{n \rightarrow \infty} \xi_n < \infty\}$ in this way, there is a more elegant argument based on the strong law of large numbers. What does the law of large numbers tell us about the limit of $\frac{\xi_n}{n}$ as $n \rightarrow \infty$?

EXERCISE 3.15. Show that ξ_n has absolutely continuous distribution with density

$$(40) \quad f_n(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}$$

with parameters n and λ . The density $f_n(t)$ of the gamma distribution is shown in ?? for $n = 2, 4$ and $\lambda = 1$.

FIGURE 4. Density $f_n(t)$ of the gamma distribution with parameters $n = 2, \lambda = 1$ and $n = 4, \lambda = 1$

Hint: Use the formula for $P\{\xi_n > t\}$ in the proof of ?? to find the distribution function of ξ_n . Is this function differentiable? What is the derivative?

EXERCISE 3.16. Prove that

$$\lim_{t \rightarrow \infty} N(t) = \infty \quad \text{a.s.}$$

Hint: What is the limit of $P\{N(k) \geq n\}$ as $k \rightarrow \infty$? Can you express $\{\lim_{t \rightarrow \infty} N(t) = \infty\}$ in terms of the sets $\{N(k) \geq n\}$?

EXERCISE 3.17. Verify that

$$P\{N(t) \text{ is odd}\} = e^{-\lambda t} \sinh(\lambda t),$$

$$P\{N(t) \text{ is even}\} = e^{-\lambda t} \cosh(\lambda t).$$

Hint: What is the probability that $N(t) = 2n + 1$? Compare this to the n -th term of the Taylor expansion of $\sinh(\lambda t)$.

EXERCISE 3.18. Show that

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \lambda \quad \text{a.s.}$$

if $N(t)$ is a Poisson process with parameter λ .

Hint: $N(n)$ is the sum of independent identically distributed random variables $N(1), N(2) - N(1), \dots, N(n) - N(n-1)$, so the strong law of large numbers can be applied to obtain the

limit of $N(n)/n$ as $n \rightarrow \infty$. Because $N(t)$ is non-decreasing, the limit will not be affected if n is replaced by a continuous parameter $t \geq 0$.

3. Brownian Motion

Imagine a cloud of smoke in completely still air. In time, the cloud will spread over a large volume, the concentration of smoke varying in a smooth manner. However, if a single smoke particle is observed, its path turns out to be extremely rough due to frequent collisions with other particles. This exemplifies two aspects of the same phenomenon called diffusion: erratic particle trajectories at the microscopic level, giving rise to a very smooth behaviour of the density of the whole ensemble of particles. The Wiener process $W(t)$ defined below is a mathematical device designed as a model of the motion of individual diffusing particles. In particular, its paths exhibit similar erratic behaviour to the trajectories of real smoke particles. Meanwhile, the density $f_{W(t)}$ of the random variable $W(t)$ is very smooth, given by the exponential function

$$(41) \quad f_{W(t)}(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}},$$

which is a solution of the diffusion equation

$$(42) \quad \frac{\partial f}{\partial t} = \frac{1}{2} \frac{\partial^2 f}{\partial x^2}$$

and can be interpreted as the density at time t of a cloud of smoke issuing from a single point source at time 0. The Wiener process $W(t)$ is also associated with the name of the British botanist Robert Brown, who around 1827 observed the random movement of pollen particles in water. We shall study mainly the one-dimensional Wiener process, which can be thought of as the projection of the position of a smoke particle onto one of the axes of a coordinate system.

Apart from describing the motion of diffusing particles, the Wiener process is widely applied in mathematical models involving various noisy systems, for example, the behaviour of asset prices at the stock exchange. If the noise in the system is due to a multitude of independent random changes, then the Central Limit Theorem predicts that the net result will have the normal distribution, a property shared by the increments $W(t) - W(s)$ of the Wiener process. This is one of the main reasons of the widespread use of $W(t)$ in mathematical models.

3.1. Definition and Basic Properties.

DEFINITION 3.12 (Wiener Process / Brownian Motion). *The Wiener process (or Brownian motion) is a stochastic process $W(t)$ with values in $\mathbb{R}$ defined for $t \in [0, \infty)$ such that*

- (1) $W(0) = 0$ a.s.;
- (2) the sample paths $t \mapsto W(t)$ are a.s. continuous;
- (3) for any finite sequence of times $0 < t_1 < \dots < t_n$ and Borel sets $A_1, \dots, A_n \subset \mathbb{R}$

$$\begin{aligned} & P\{W(t_1) \in A_1, \dots, W(t_n) \in A_n\} \\ &= \int_{A_1} \dots \int_{A_n} p(t_1, 0, x_1) p(t_2 - t_1, x_1, x_2) \dots \\ & \quad \dots p(t_n - t_{n-1}, x_{n-1}, x_n) dx_1 \dots dx_n, \end{aligned}$$

where

$$(6.5) \quad p(t, x, y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}}$$

defined for any $x, y \in \mathbb{R}$ and $t > 0$ is called the transition density.

A typical sample path of the Wiener process is shown in ??.

FIGURE 5. A typical path of $W(t)$

EXERCISE 3.19. Show that

$$f_{W(t)}(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$$

is the probability density of $W(t)$ and find the expectation and variance of $W(t)$.

Hint: The density of $W(t)$ can be obtained from condition 3) of ?? written for a single time t and a single Borel set. You will need the formula

$$\int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} dx = \sqrt{2\pi}$$

to compute the integrals in the expressions for the expectation and variance.

REMARK 3.1. The results of Exercise ?? mean that $W(t)$ has the normal distribution with mean 0 and variance t .

EXERCISE 3.20. Show that

$$E(W(s)W(t)) = \min\{s, t\}.$$

Hint: The joint density of $W(s)$ and $W(t)$ will be needed. It can be found from condition 3) of ?? written for two times s and t and two Borel sets.

EXERCISE 3.21. Show that

$$E(|W(t) - W(s)|^2) = |t - s|.$$

Hint: Expand the square and use the formula in Exercise ??.

EXERCISE 3.22. Compute the characteristic function $E(\exp(i\lambda W(t)))$ for any $\lambda \in \mathbb{R}$.

Hint: Use the density of $W(t)$ found in Exercise ??.

EXERCISE 3.23. Find $E(W(t)^4)$.

Hint: This can be done, for example, by expressing the expectation in terms of the density of $W(t)$ and computing the resulting integral, or by computing the fourth derivative of the characteristic function of $W(t)$ at 0. The second method is more efficient.

DEFINITION 3.13 (n -dimensional Wiener Process). We call $W(t) = (W^1(t), \dots, W^n(t))$ an n -dimensional Wiener process if $W^1(t), \dots, W^n(t)$ are independent $\mathbb{R}$ -valued Wiener processes.

EXERCISE 3.24. For a two-dimensional Wiener process $W(t) = (W^1(t), W^2(t))$ find the probability that $|W(t)| < R$, where $R > 0$ and $|x|$ is the Euclidean norm of $x = (x^1, x^2)$ in $\mathbb{R}^2$, i.e. $|x|^2 = (x^1)^2 + (x^2)^2$.

Hint: Express the probability in terms of the joint density of $W^1(t)$ and $W^2(t)$. Independence means that the joint density of $W^1(t)$ and $W^2(t)$ is the product of their respective densities, which are known from Exercise ???. It is convenient to use polar coordinates to compute the resulting integral over a disc.

3.2. Increments of Brownian Motion.

Proposition 3.14 (Increments are Normal). *For any $0 \leq s < t$ the increment $W(t) - W(s)$ has the normal distribution with mean 0 and variance $t - s$.*

PROOF. By condition 3) of ?? the joint density of $W(s), W(t)$ is

$$f_{W(s), W(t)}(x, y) = p(s, 0, x)p(t - s, x, y).$$

Hence, for any Borel set A

$$\begin{aligned}
 P\{W(t) - W(s) \in A\} &= \int_{\{(x,y): y-x \in A\}} p(s, 0, x)p(t - s, x, y) dx dy \\
 &= \int_{-\infty}^{+\infty} p(s, 0, x) \left(\int_{\{y: y-x \in A\}} p(t - s, x, y) dy \right) dx \\
 &= \int_{-\infty}^{+\infty} p(s, 0, x) \left(\int_A p(t - s, x, x + u) du \right) dx \\
 (43) \qquad &= \int_{-\infty}^{+\infty} p(s, 0, x) \left(\int_A p(t - s, 0, u) du \right) dx \\
 &= \int_A p(t - s, 0, u) du \int_{-\infty}^{+\infty} p(s, 0, x) dx \\
 &= \int_A p(t - s, 0, u) du.
 \end{aligned}$$

But $f(u) = p(t - s, 0, u)$ is the density of the normal distribution with mean 0 and variance $t - s$, which proves the claim.

Corollary 3.15. ?? implies that $W(t)$ has stationary increments.

Proposition 3.16 (Independent Increments). *For any $0 = t_0 \leq t_1 \leq \dots \leq t_n$ the increments*

$$W(t_1) - W(t_0), \dots, W(t_n) - W(t_{n-1})$$

are independent.

PROOF. From ?? we know that the increments of $W(t)$ have the normal distribution. Because normally distributed random variables are independent if and only if they are uncorrelated, it suffices to verify that

$$E[(W(u) - W(t))(W(s) - W(r))] = 0$$

for any $0 \leq r \leq s \leq t \leq u$. But this follows immediately from Exercise ??:

$$\begin{aligned}
 E[(W(u) - W(t))(W(s) - W(r))] &= E(W(u)W(s)) - E(W(u)W(r)) \\
 &\quad - E(W(t)W(s)) + E(W(t)W(r)) \\
 (44) \qquad &= s - r - s + r \\
 &= 0,
 \end{aligned}$$

as required.

Corollary 3.17. For any $0 \leq s < t$ the increment $W(t) - W(s)$ is independent of the σ -field $\mathcal{F}_s = \sigma\{W(r) : 0 \leq r \leq s\}$.

PROOF. By ?? the random variables $W(t) - W(s)$ and $W(r) - W(0) = W(r)$ are independent if $0 \leq r \leq s \leq t$. Because the σ -field $\mathcal{F}_s$ is generated by such $W(r)$, it follows that $W(t) - W(s)$ is independent of $\mathcal{F}_s$.

EXERCISE 3.25. Show that $W(t)$ is a martingale with respect to the filtration $\mathcal{F}_t$.

Hint: Take advantage of the fact that $W(t) - W(s)$ is independent of $\mathcal{F}_s$ if $s < t$.

EXERCISE 3.26. Show that $|W(t)|^2 - t$ is a martingale with respect to the filtration $\mathcal{F}_t$.

Hint: Once again, use the fact that $W(t) - W(s)$ is independent of $\mathcal{F}_s$ if $s < t$.

Let us state without proof the following useful characterization of the Wiener process in terms of its increments.

Theorem 3.18 (Wiener Process Characterization). A stochastic process $W(t), t \geq 0$, is a Wiener process if and only if the following conditions hold:

- (1) $W(0) = 0$ a.s.;
- (2) the sample paths $t \mapsto W(t)$ are continuous a.s.;
- (3) $W(t)$ has stationary independent increments;
- (4) the increment $W(t) - W(s)$ has the normal distribution with mean 0 and variance $t - s$ for any $0 \leq s < t$.

EXERCISE 3.27. Show that for any $T > 0$

$$V(t) = W(t + T) - W(T)$$

is a Wiener process if $W(t)$ is.

Hint: Are the increments of $V(t)$ independent? What is their distribution? Does $V(t)$ have continuous paths? Is it true that $V(0) = 0$?

The Wiener process can also be characterized by its martingale properties. The following theorem is also given without proof.

Theorem 3.19 (Lévy's Martingale Characterization). Let $W(t), t \geq 0$, be a stochastic process and let $\mathcal{F}_t = \sigma(W_s, s \leq t)$ be the filtration generated by it. Then $W(t)$ is a Wiener process if and only if the following conditions hold:

- (1) $W(0) = 0$ a.s.;
- (2) the sample paths $t \mapsto W(t)$ are continuous a.s.;
- (3) $W(t)$ is a martingale with respect to the filtration $\mathcal{F}_t$;
- (4) $|W(t)|^2 - t$ is a martingale with respect to $\mathcal{F}_t$.

EXERCISE 3.28. Let $c > 0$. Show that $V(t) = \frac{1}{c}W(c^2t)$ is a Wiener process if $W(t)$ is.

Hint: Is $V(t)$ a martingale? With respect to which filtration? Is $|V(t)|^2 - t$ a martingale? Are the paths of $V(t)$ continuous? Is it true that $V(0) = 0$?

3.3. Sample Paths. Let

$$0 = t_0^n < t_1^n < \cdots < t_n^n = T,$$

where

$$t_i^n = \frac{iT}{n},$$

be a partition of the interval $[0, T]$ into n equal parts. We denote by

$$\Delta_i^n W = W(t_{i+1}^n) - W(t_i^n)$$

the corresponding increments of the Wiener process $W(t)$.

EXERCISE 3.29. Show that

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} (\Delta_i^n W)^2 = T \quad \text{in } L^2.$$

Hint: You need to show that

$$\lim_{n \rightarrow \infty} E \left(\left[\sum_{i=0}^{n-1} (\Delta_i^n W)^2 - T \right]^2 \right) = 0.$$

Use the independence of increments to simplify the expectation. What are the expectations of $\Delta_i^n W$, $(\Delta_i^n W)^2$ and $(\Delta_i^n W)^4$?

The next theorem on the variation of the paths of $W(t)$ is a consequence of the result in Exercise ???. First, let us recall that the variation of a function is defined as follows.

DEFINITION 3.20 (Variation). The variation of a function $f : [0, T] \rightarrow \mathbb{R}$ is defined to be

$$\limsup_{\Delta t \rightarrow 0} \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)|,$$

where $t = (t_0, t_1, \dots, t_n)$ is a partition of $[0, T]$, i.e. $0 = t_0 < t_1 < \dots < t_n = T$, and where

$$\Delta t = \max_{i=0, \dots, n-1} |t_{i+1} - t_i|.$$

Theorem 3.21 (Infinite Variation). The variation of the paths of $W(t)$ is infinite a.s.

PROOF. Consider the sequence of partitions $t^n = (t_0^n, t_1^n, \dots, t_n^n)$ of $[0, T]$ into n equal parts. Then

$$\sum_{i=0}^{n-1} |\Delta_i^n W|^2 \leq \left(\max_{i=0, \dots, n-1} |\Delta_i^n W| \right) \sum_{i=0}^{n-1} |\Delta_i^n W|.$$

Since the paths of $W(t)$ are a.s. continuous on $[0, T]$,

$$\lim_{n \rightarrow \infty} \left(\max_{i=0, \dots, n-1} |\Delta_i^n W| \right) = 0 \quad \text{a.s.}$$

By Exercise ?? there is a subsequence $t^{n_k} = (t_0^{n_k}, t_1^{n_k}, \dots, t_{n_k}^{n_k})$ of partitions such that

$$\lim_{k \rightarrow \infty} \sum_{i=0}^{n_k-1} |\Delta_i^{n_k} W|^2 = T \quad \text{a.s.}$$

This is because every sequence of random variables convergent in L^2 has a subsequence convergent a.s. It follows that

$$\lim_{k \rightarrow \infty} \sum_{i=0}^{n_k-1} |\Delta_i^{n_k} W| = \infty \quad \text{a.s.},$$